



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

STUDENT SKILL DEVELOPMENT MANAGEMENT SYSTEM

ASSIGNMENT REPORT

CSA4305 – INTERNET PROGRAMMING

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING**

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1. INTRODUCTION

The Student Skill Development Management System is a web-based application designed to manage, evaluate, and improve the overall skills and competencies of students. In a modern educational environment, academic performance alone is not sufficient to determine whether a student is ready for higher studies, internships, or employment. Students are expected to possess technical knowledge, programming skills, communication abilities, problem-solving skills, teamwork, certifications, project experience, internship experience, and other professional competencies. However, these details are often maintained separately, making it difficult to obtain a complete picture of a student's development.

The proposed Student Skill Development Management System provides a centralized platform where students can create their skill profiles and maintain information about their certifications, projects, workshops, internships, and other achievements. The system also provides skill assessments through which the current competency of a student can be measured. Assessment results can be compared with expected competency levels to identify skill gaps. Based on the identified gaps, suitable learning activities and courses can be recommended to students.

The system also provides facilities for faculty members and mentors to monitor student development and provide feedback. Administrators can manage student records and generate analytical reports. Therefore, the proposed system provides a common platform connecting students, faculty members, mentors, and administrators.

2. PROBLEM ANALYSIS

Many educational institutions maintain academic information such as marks, attendance, and course results in one system while skill-related information is maintained separately or manually. Programming skills, communication skills, certifications, projects, workshops, internships, and other competencies may not be available in one centralized record. This makes it difficult for students to understand their complete skill profile.

Students may know that they need to improve a particular skill but may not know their exact competency gap or which learning activity is most suitable. Faculty members and mentors also face difficulties when they need to evaluate a student's development using information collected from different sources. Administrators may not have sufficient analytical information to determine the overall skill requirements of students.

The proposed system solves this problem by maintaining a centralized database of student development information. The system can collect skill information, assessment scores, certifications, project details, learning activities, and mentor feedback. It can then use this information to provide a more complete representation of the student's competency.

3. OBJECTIVES

The primary objective of the Student Skill Development Management System is to provide an integrated platform for managing and monitoring student skill development. The system is designed to allow students to create their profiles, add their skills, record certifications and projects, participate in assessments, identify competency gaps, access learning recommendations, and monitor their progress.

Another objective is to provide faculty members and mentors with a facility to review student development and provide feedback. The system also aims to provide administrators with analytical information that can be used for decision making and employability planning. The system additionally aims to protect personal information by providing controlled access to student records.

4. PROPOSED SYSTEM

The proposed system is divided into several interconnected functional components. The student profile component manages personal and academic information. The skill management component stores technical and non-technical skills along with competency levels. The certification and project component maintains details of completed certifications and projects.

The assessment component evaluates student competency through tests or other evaluation methods. After assessment, the system compares the current competency level with the required competency level. If the current level is lower than the required level, the difference is treated as a competency gap.

The learning recommendation component suggests suitable courses, workshops, practice activities, and training programs based on the identified gap. The progress monitoring component stores previous and current assessment results so that improvement can be measured. The mentor feedback component allows faculty members or mentors to provide comments and recommendations. The reporting component generates useful reports for students, mentors, faculty members, and administrators.

5. SYSTEM WORKFLOW

The system begins with student registration and authentication. After successful login, the student creates a profile containing the required information. The student can then add different skills and specify the current competency level for each skill.

The student can also add information about certifications, projects, workshops, and internships. After completing the profile, the student can take skill assessments. The assessment result is processed by the system and the competency level is calculated.

The system then compares the student's competency with the required competency. If the student has achieved the required level, the system indicates that no major competency gap exists. If the competency is below the expected level, the system identifies the gap and recommends suitable learning activities.

After completing recommended activities, the student can update the progress information and take another assessment. The new result can be compared with the previous result to determine improvement. Faculty members or mentors can review this information and provide feedback.

6. SYSTEM MODULES

6.1 Student Profile Module

The Student Profile Module is responsible for maintaining the basic information of each student. It can contain information such as student ID, name, department, email address, year of study, and other relevant details. The profile provides the foundation for storing all other skill-development information.

6.2 Skill Management Module

The Skill Management Module allows students to add and update their technical and non-technical skills. Skills can be categorized into programming, communication, problem solving, leadership, teamwork, database management, web development, and other areas. Each skill can have a competency level such as beginner, intermediate, advanced, or expert.

6.3 Certification Module

The Certification Module allows students to record certificates obtained from academic institutions, online learning platforms, workshops, or other recognized sources. Certification information may include certificate name, issuing organization, completion date, and verification details.

6.4 Project Management Module

The Project Module maintains information about student projects. Students can provide the project title, description, technologies used, role in the project, duration, and project outcome. Project records help faculty members and recruiters understand practical experience.

6.5 Skill Assessment Module

The Skill Assessment Module evaluates the competency of students. Assessment questions can be related to a particular skill. The system calculates the score and stores the result. The score can then be mapped to a competency level.

6.6 Competency Gap Module

The Competency Gap Module compares the current competency level of a student with the required competency level. If the current level is lower than the required level, the system calculates the difference and identifies the skill as an area requiring improvement.

6.7 Learning Recommendation Module

The Learning Recommendation Module uses identified competency gaps to recommend suitable learning activities. For example, if a student's JavaScript skill level is below the expected level, the system may recommend JavaScript practice, web-development courses, or relevant workshops.

6.8 Progress Monitoring Module

The Progress Monitoring Module records changes in competency over time. Students can compare previous and current scores and understand their improvement. Mentors can also use this information to provide more effective guidance.

6.9 Faculty and Mentor Module

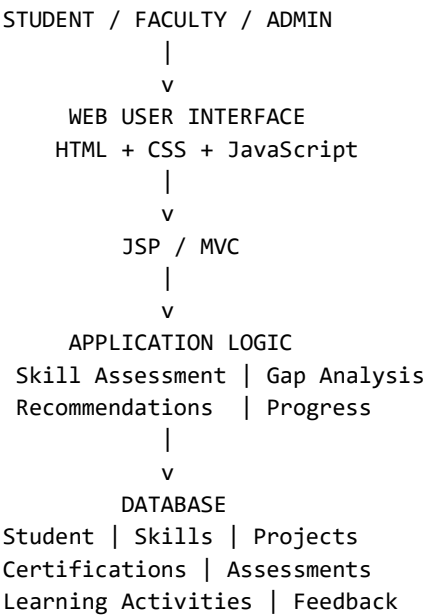
Faculty members and mentors can view student development information according to their access permissions. They can review assessments, projects, certifications, and progress and provide feedback for improvement.

6.10 Administration and Reporting Module

The administrator manages users, skills, learning activities, and system records. Analytical reports can be generated to identify common skill gaps, student development trends, and overall institutional performance.

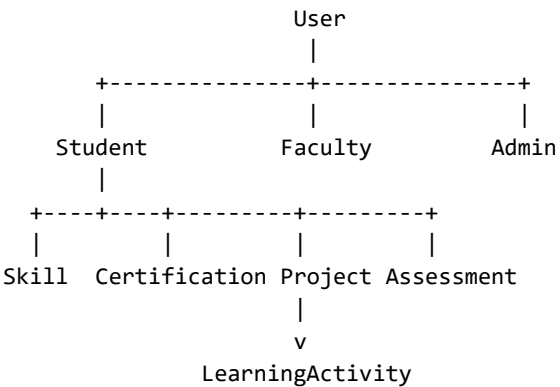
7. SYSTEM ARCHITECTURE

The proposed system follows a layered architecture. The presentation layer provides the user interface through HTML, CSS, JavaScript, and JSP. The application layer contains business logic for skill evaluation, competency-gap identification, recommendations, and progress calculations. The data layer manages student, skill, assessment, certification, project, and progress information.



8. OBJECT-ORIENTED CLASS DESIGN

The major entities in the system can be represented using classes. A Student class represents student information, while a Skill class represents an individual competency. Certification and Project classes represent student achievements. Assessment represents an evaluation conducted for a particular skill. LearningActivity represents recommended learning content, while Mentor represents faculty or mentor information.



9. SAMPLE STUDENT CLASS

The following Java code demonstrates a simple Student class that can be used as part of the backend implementation.

```
class Student {

    private int studentId;
    private String name;
    private String department;
    private String email;

    static int studentCount = 0;

    Student(int studentId, String name,
            String department, String email) {

        this.studentId = studentId;
        this.name = name;
        this.department = department;
        this.email = email;

        studentCount++;
    }

    public void displayStudent() {
        System.out.println("Student ID : " + studentId);
        System.out.println("Name      : " + name);
        System.out.println("Department : " + department);
        System.out.println("Email    : " + email);
    }

    public static int getStudentCount() {
        return studentCount;
    }
}
```

10. SKILL CLASS

The Skill class represents a student's competency in a particular area and provides the basic logic required for competency-gap analysis.

```
class Skill {

    private String skillName;
    private String category;
    private int currentLevel;
    private int requiredLevel;

    Skill(String skillName, String category,
          int currentLevel, int requiredLevel) {

        this.skillName = skillName;
        this.category = category;
        this.currentLevel = currentLevel;
        this.requiredLevel = requiredLevel;
    }
}
```

```

    public int calculateGap() {
        if (currentLevel >= requiredLevel)
            return 0;

        return requiredLevel - currentLevel;
    }

    public void displaySkill() {
        System.out.println("Skill : " + skillName);
        System.out.println("Category : " + category);
        System.out.println("Current Level : " + currentLevel);
        System.out.println("Required Level : " + requiredLevel);
        System.out.println("Skill Gap : " + calculateGap());
    }
}

```

11. INHERITANCE AND POLYMORPHISM

The system can use inheritance by creating a general User class and deriving Student, Faculty, and Administrator classes from it. Common information such as name and email can be maintained in the parent class, while specialized behavior can be implemented in child classes.

```

class User {

    protected String name;
    protected String email;

    User(String name, String email) {
        this.name = name;
        this.email = email;
    }

    void displayRole() {
        System.out.println("General User");
    }
}

class Student extends User {

    Student(String name, String email) {
        super(name, email);
    }

    @Override
    void displayRole() {
        System.out.println("Student");
    }
}

class Faculty extends User {

    Faculty(String name, String email) {
        super(name, email);
    }
}

```

```

    @Override
    void displayRole() {
        System.out.println("Faculty");
    }
}

```

12. COMPETENCY GAP ANALYSIS CODE

The following JavaScript code can be used in the web application to calculate the competency gap automatically.

```

function calculateGap() {

    let current =
        Number(document.getElementById("currentLevel").value);

    let required =
        Number(document.getElementById("requiredLevel").value);

    let result =
        document.getElementById("result");

    if (current < 1 || current > 5 ||
        required < 1 || required > 5) {

        result.innerHTML =
            "Enter skill levels between 1 and 5.";

        return;
    }

    if (current >= required) {

        result.innerHTML =
            "No competency gap identified.";

    } else {

        let gap = required - current;

        result.innerHTML =
            "Competency Gap = " + gap +
            "<br>Recommended Action: Complete additional learning activities.";

    }

}

```

13. HTML IMPLEMENTATION

```

<!DOCTYPE html>
<html>
<head>
    <title>Student Skill Development System</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

```

```

<div class="container">

    <h1>Student Skill Development Management System</h1>

    <h2>Skill Gap Analysis</h2>

    <label>Student Name</label>
    <input type="text" id="studentName">

    <label>Skill Name</label>
    <input type="text" id="skillName">

    <label>Current Skill Level</label>
    <input type="number" id="currentLevel" min="1" max="5">

    <label>Required Skill Level</label>
    <input type="number" id="requiredLevel" min="1" max="5">

    <button onclick="calculateGap()">
        Analyze Skill Gap
    </button>

    <div id="result"></div>

</div>

<script src="script.js"></script>

</body>
</html>

```

14. CSS IMPLEMENTATION

```

body {
    font-family: Arial, sans-serif;
    background: #f4f6f8;
    margin: 0;
    padding: 40px;
}

.container {
    max-width: 700px;
    margin: auto;
    background: white;
    padding: 30px;
    border-radius: 12px;
}

h1 {
    text-align: center;
}

label {
    display: block;
    margin-top: 15px;
}

```

```

input {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
}

button {
    margin-top: 20px;
    padding: 12px 20px;
    cursor: pointer;
}

#result {
    margin-top: 20px;
    padding: 15px;
    border-radius: 8px;
}

```

15. XML DATA REPRESENTATION

XML can be used to represent structured student skill information. The hierarchical structure makes it suitable for storing and transferring student profile and competency information.

```

<?xml version="1.0" encoding="UTF-8"?>

<student>
    <studentId>101</studentId>
    <name>Reshwanth</name>
    <department>CSE</department>

    <skills>
        <skill>
            <name>Java</name>
            <category>Programming</category>
            <currentLevel>3</currentLevel>
            <requiredLevel>5</requiredLevel>
        </skill>

        <skill>
            <name>HTML</name>
            <category>Web Development</category>
            <currentLevel>4</currentLevel>
            <requiredLevel>5</requiredLevel>
        </skill>
    </skills>
</student>

```

16. XSLT TRANSFORMATION

XSLT can transform XML information into an HTML presentation so that structured student skill information can be displayed in a readable report.

```

<?xml version="1.0" encoding="UTF-8"?>

<xsl:stylesheet
version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

```

```

<xsl:template match="/">

<html>
<body>

<h2>Student Skill Profile</h2>

<table border="1">

<tr>
<th>Skill</th>
<th>Category</th>
<th>Current Level</th>
<th>Required Level</th>
</tr>

<xsl:for-each select="student/skills/skill">

<tr>
<td><xsl:value-of select="name"/></td>
<td><xsl:value-of select="category"/></td>
<td><xsl:value-of select="currentLevel"/></td>
<td><xsl:value-of select="requiredLevel"/></td>
</tr>

</xsl:for-each>

</table>

</body>
</html>

</xsl:template>
</xsl:stylesheet>

```

17. JSP IMPLEMENTATION

JSP can be used to display dynamic student information received from a request. The following example demonstrates a basic JSP page.

```

<%@ page language="java"
      contentType="text/html; charset=UTF-8" %>

<!DOCTYPE html>
<html>
<head>
<title>Student Profile</title>
</head>

<body>

<h1>Student Skill Profile</h1>

<%
String name = request.getParameter("name");
String department = request.getParameter("department");

```

```

if(name == null)
    name = "Student";

if(department == null)
    department = "CSE";
%>

<p>Student Name: <%= name %></p>
<p>Department: <%= department %></p>

</body>
</html>

```

18. MVC DESIGN

The MVC design divides the application into Model, View, and Controller components. The Model contains the data and business logic. For this project, Student, Skill, Certification, Project, Assessment, and LearningActivity can act as model components.

The View is responsible for presenting information to the user. JSP pages can be used as views. The Controller receives requests from users and determines which operation should be performed. For example, when a student submits a skill assessment, the controller receives the request, sends the information to the model for processing, and then returns the result to the appropriate JSP page.

19. PSEUDOCODE

The system starts by displaying the login page and authenticating the user. After successful authentication, the system identifies whether the user is a student, faculty member, mentor, or administrator. If the user is a student, the system displays the student dashboard and allows the student to create or update the skill profile. The student can add skills, certifications, projects, workshops, and internships. The student can then take an assessment for a selected skill. The system calculates the assessment score and determines the competency level. The current competency is compared with the required competency. If a gap exists, the system recommends suitable learning activities. After completing the learning activities, the student updates the progress information and can take another assessment. The system compares the new result with the previous result and generates a progress report. Faculty members and mentors can review the information and provide feedback, while administrators can generate analytical reports.

20. SAMPLE INPUT

Student Name	: Reshwanth
Student ID	: 101
Department	: CSE
Skill	: Java
Current Level	: 3
Required Level	: 5

21. SAMPLE OUTPUT

Student Skill Development Management System

Student Name : Reshwanth

Skill : Java

Current Level : 3

Required Level : 5

Competency Gap : 2

Recommendation:

Complete Java programming practice,
advanced Java course and project-based learning.

If the current competency is equal to or greater than the required level, the system displays that the required competency has been achieved and no major skill gap has been identified.

22. DATABASE DESIGN

The database can contain separate tables for Student, Skill, Certification, Project, Assessment, LearningActivity, Progress, MentorFeedback, and User. The Student table stores student identification and profile information. The Skill table stores available skills and categories. A StudentSkill table can connect students with their selected skills and competency levels.

The Certification table stores certificate details, while the Project table stores project information. The Assessment table stores assessment scores and dates. The LearningActivity table stores courses, workshops, and training activities. The Progress table records changes in competency over time. MentorFeedback stores comments provided by mentors or faculty members.

23. SAMPLE SQL TABLE

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    department VARCHAR(100),  
    email VARCHAR(100)  
);  
  
CREATE TABLE Skill (  
    skill_id INT PRIMARY KEY,  
    skill_name VARCHAR(100),  
    category VARCHAR(100)  
);  
  
CREATE TABLE StudentSkill (  
    student_id INT,  
    skill_id INT,  
    current_level INT,  
    required_level INT,  
  
    PRIMARY KEY(student_id, skill_id),  
  
    FOREIGN KEY(student_id)  
    REFERENCES Student(student_id),  
  
    FOREIGN KEY(skill_id)  
    REFERENCES Skill(skill_id)  
);
```

24. TESTING

Testing is performed to verify that the system behaves correctly under different conditions. Login testing checks whether valid users can access the system and invalid credentials are rejected. Profile testing checks whether student information can be added, updated, and displayed correctly. Skill testing checks whether valid competency levels are accepted.

Assessment testing verifies that scores are calculated correctly. Competency-gap testing checks whether the difference between current and required levels is correctly identified. Recommendation testing verifies that suitable learning activities are displayed for identified gaps. Progress testing verifies whether new assessment results are correctly compared with previous results.

Input validation testing is also important. The system should reject invalid skill levels, empty required fields, incorrect email formats, and other invalid data. Security testing should verify that students cannot access restricted administrative information.

25. EXPECTED RESULTS

After successful implementation, students will be able to maintain a complete digital skill profile. They will be able to record certifications and projects, perform assessments, identify competency gaps, receive learning recommendations, and monitor progress.

Faculty members and mentors will be able to review student development and provide feedback. Administrators will be able to monitor overall student skill development and generate analytical reports. The system will therefore provide a unified environment for student skill development management.

26. ADVANTAGES

The proposed system provides a centralized platform for student skill information and reduces the dependency on scattered records. It helps students understand their strengths and weaknesses and provides recommendations based on identified competency gaps. It also allows mentors to provide data-based feedback and helps institutions analyze overall student development.

The system improves transparency because students can monitor their own progress. It can also support employability by maintaining information about certifications, projects, internships, assessments, and other competencies in one location.

27. LIMITATIONS

The accuracy of recommendations depends on the quality of the skill data and assessment results available in the system. If students enter incorrect information, the generated profile may not accurately represent their competency. The system also requires regular maintenance of learning resources and skill requirements.

Automated assessment cannot always measure complex skills such as creativity, leadership, communication quality, or teamwork accurately. Therefore, mentor and faculty feedback remains important for comprehensive evaluation.

28. FUTURE ENHANCEMENTS

The system can be enhanced by integrating artificial intelligence and machine-learning techniques to provide personalized learning recommendations. An AI-based recommendation engine could analyze previous assessment results, learning behavior, project experience, and career interests to suggest suitable courses and career paths.

The system can also be integrated with online learning platforms so that completed courses can automatically update student profiles. Resume generation and placement-management features can be added to connect student competencies with job requirements. Mobile application support can also be introduced to allow students to monitor their skill development from smartphones.

Another future enhancement is the development of advanced dashboards containing charts and graphs. These dashboards can help students and administrators understand skill development trends more easily.

29. SDG RELEVANCE

The project is strongly related to Sustainable Development Goal 4 because it supports quality education and continuous learning. By identifying competency gaps and recommending suitable learning activities, the system encourages students to continuously improve their knowledge and skills.

The project also supports SDG 8 by improving employability and helping students develop skills required for employment. Certifications, projects, internships, and assessment results provide evidence of student competency.

The system is also related to SDG 9 because it uses digital technology to improve educational processes and institutional decision making.

30. DESIGN DECISIONS AND REFLECTION

The centralized architecture was selected because it provides a single source of information for student skill development. Object-oriented design was selected because the major system entities can naturally be represented as objects. Encapsulation provides better control over student data, inheritance reduces repeated implementation, and polymorphism provides flexibility for different user roles.

The use of HTML and CSS provides the basic structure and visual design of the application. JavaScript provides dynamic interaction and validation. XML provides structured data representation, while XSLT provides transformation and presentation of XML data. JSP and MVC provide dynamic web application functionality and separate the user interface from application processing.

One implementation consideration is that Java does not directly support traditional operator overloading. Therefore, if the system is implemented using Java/JSP, comparison operations can be implemented using methods such as `equals()`, `compareTo()`, or custom comparison methods. If the rubric specifically requires actual operator overloading syntax, a language such as C++ would be required for that particular requirement.

31. CONCLUSION

The Student Skill Development Management System provides a practical solution for managing the complete skill-development journey of students. Instead of maintaining academic records, certifications, projects, assessments, and competency information separately, the proposed system combines these components into a centralized platform.

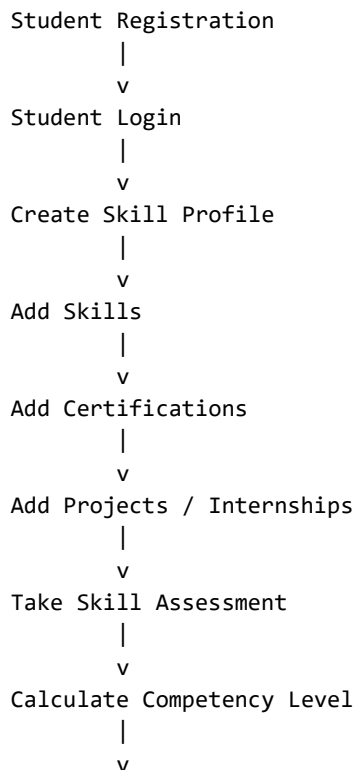
The system enables students to create skill profiles, record achievements, perform assessments, identify competency gaps, access learning recommendations, and monitor their progress. Faculty members and mentors can review student development and provide feedback, while administrators can generate reports and analyze institutional skill-development requirements.

The implementation demonstrates important Internet Programming concepts including HTML, CSS, JavaScript, XML, XSLT, JSP, and MVC. The use of object-oriented design concepts further improves the structure and maintainability of the application. The project also supports SDG 4, SDG 8, and SDG 9 by promoting quality education, skill development, employability, and technology-driven educational management.

32. GITHUB UPLOAD

The complete project can be organized into separate folders for frontend, backend, XML/XSLT files, JSP pages, database scripts, documentation, and screenshots. The source code and project documentation can then be uploaded to GitHub. The repository should contain the complete implementation along with a README file explaining the project title, objectives, technologies used, installation procedure, execution procedure, sample outputs, and future enhancements.

33. FINAL PROJECT FLOW



Identify Skill Gap

|

v

Recommend Learning Activities

|

v

Complete Learning Activities

|

v

Update Progress

|

v

Reassessment

|

v

Mentor Feedback

|

v

Generate Skill Development Report

|

v

Administrative Analysis

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Reshwanth 192411235	Requirements & Skill Module	System flow & profile module	Module testing	Introduction & Objectives	25%
Maheshwar 192411195	Frontend & Dashboard	HTML/CSS & dashboard	UI testing	UI & Modules	25%
Vishnu Vardhan 192311267	Assessment & Progress	JavaScript & assessment	Function testing	Assessment & Progress	25%
Pradeep Ranga 192311289	Backend & Database	Java, XML & SQL	Data testing	Backend & Database	25%
Total					100%